

Lab 1.1 – Hardware Set-up & Getting Connected

At the end of this lab, Participants will be able to:

- Set up the Hardware for the In-Sight system
- Connect to the In-Sight camera using In-Sight EasyBuilder

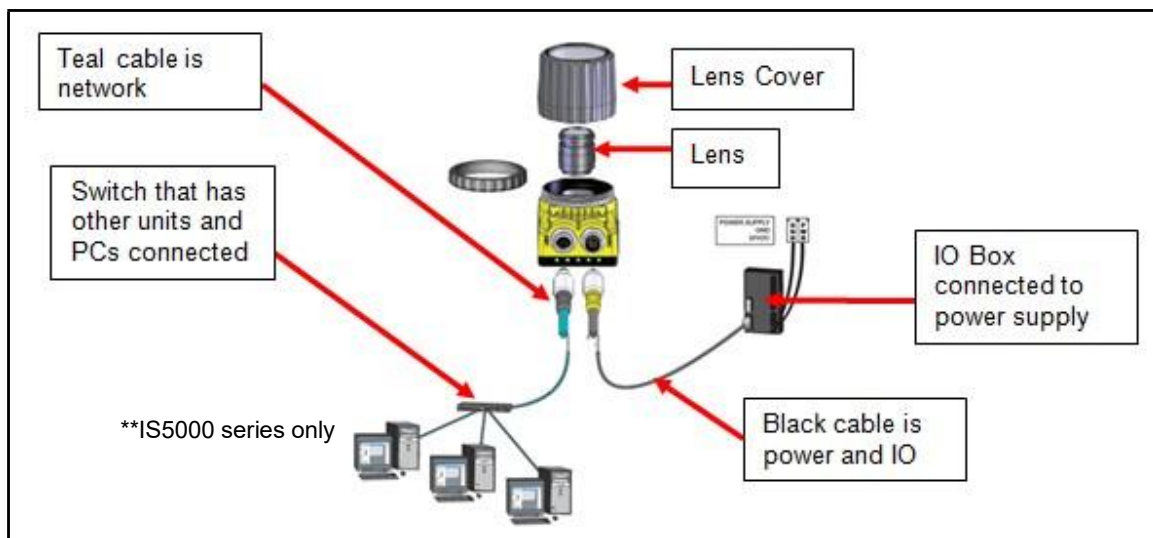
The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

- Easy Builder User Interface Application Step 1
 - Getting Connected

Follow the steps below to complete the lab (if using an In-Sight Camera):

1. Assemble the hardware components.

NOTE: The tripod should have the top three portions of its legs pushed back in to get the proper height. The unit should be directly above the part below it with the lens pointing down.



2. Confirm there is power and network connectivity to the unit.

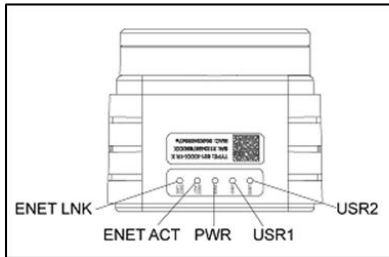
Normal LED Pattern:

7000 Series – power LED and ENET connector should be lit in green

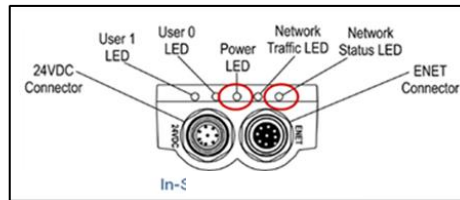
7000 Series Gen II – power LED  green, network LED  yellow

5000 Series – power LED and ENET connector should be lit in green

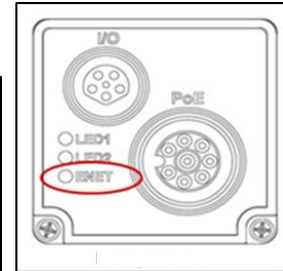
Micro Series – ENET LED should be green



In-Sight 7000 Series



In-Sight 5000 Series

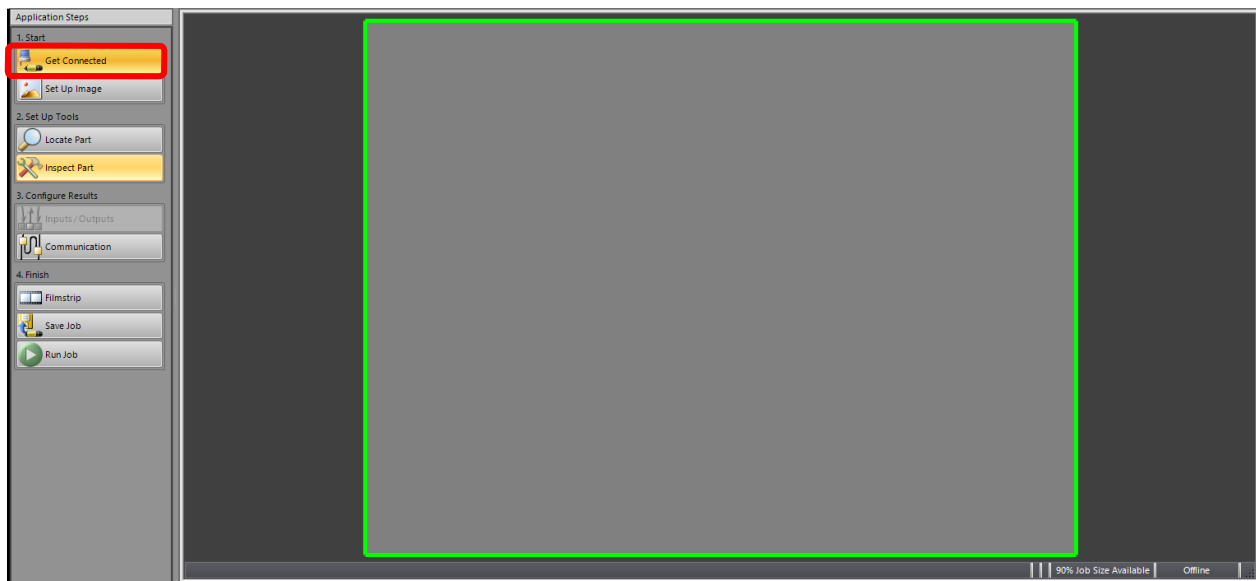


In-Sight Micro

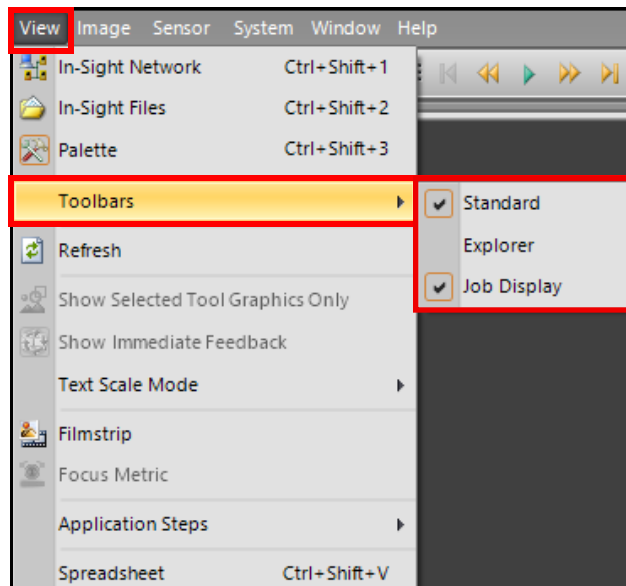


In-Sight 7000 Series Gen II

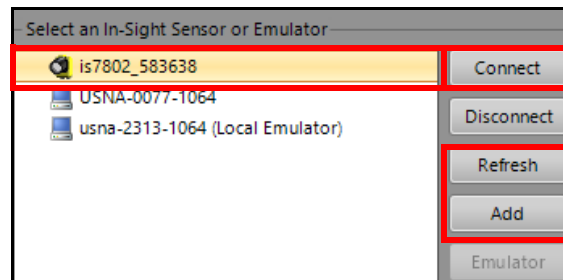
3. Click the **In-Sight Explorer** icon on your computer to launch In-Sight Explorer. Or, Start Menu → Programs → Cognex → In-Sight Explorer (ISE).
4. Confirm that you are in the **Easy Builder View** and the Application step – **Get Connected** is highlighted, if it is not click on it.



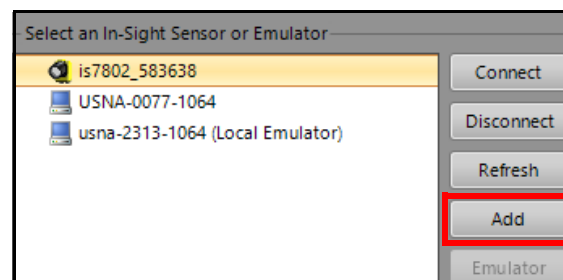
5. Click on **View -> Toolbars**, confirm that **Job Display** and **Standard** are checked.



6. In the Settings Pane, click on your camera in the list of systems. Do not select the Emulator. Skip Steps 7 to 11 and jump to Step 12 if the camera is in the list.
NOTE: The camera's default name is the model plus the last 6 digits of the camera's MAC address found on the top of the camera case. The name you see may be different, if a previous user changed it.



7. If the camera is not in the list, click the **Refresh** button. If it still does not appear, click the **Add** button and see if it displays. If not, cycle power to the camera while in the **Add** window and it should appear.
8. Next, you are going to set your camera to factory defaults. This will remove any change in settings made to your camera by a previous class. Click the **Add** button.



The **Add Sensor/Device to Network** dialog displays.

Note: Your camera's name should **not** appear as it is now properly configured.

Select an In-Sight sensor or device to add to your network. If the desired sensor/device is not listed, you can add it by cycling its power. Devices may take up to 60 seconds to appear in the list after power is applied.

Filter:

Host Name	Type	MAC	IP
-----------	------	-----	----

Host Name:

☐ Obtain IP Address Automatically (DHCP)

☒ Use The Following Network Settings

IP Address:

Subnet Mask:

Default Gateway:

DNS Server:

Domain Name:

☐ Reset Admin Password

☐ Reset Sensor Settings to Factory Defaults

☒ Show All ☐ Show New

9. To show all of the cameras that are properly networked, click the **Show All** radio button. After a few seconds, a list of all cameras on the network, including yours, should appear.
10. Click on your camera, click the **Reset Sensor Settings to Factory Defaults** checkbox and click the **Apply** button.

Select an In-Sight sensor or device to add to your network. If the desired sensor/device is not listed, you can add it by cycling its power. Devices may take up to 60 seconds to appear in the list after power is applied.

Filter:

Host Name	Type	MAC	IP
is7802_5...	7802	00-d0-24-58-36-38	169.254.0.5

Host Name:

☐ Obtain IP Address Automatically (DHCP)

☒ Use The Following Network Settings

IP Address:

Subnet Mask:

Default Gateway:

DNS Server:

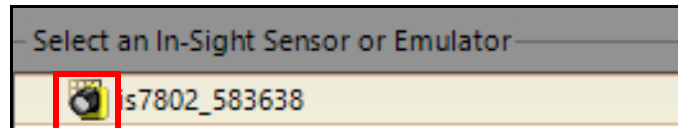
Domain Name:

☐ Reset Admin Password

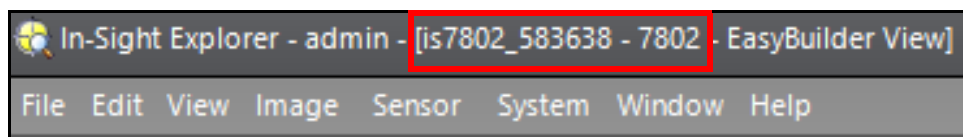
☒ Reset Sensor Settings to Factory Defaults

☒ Show All ☐ Show New

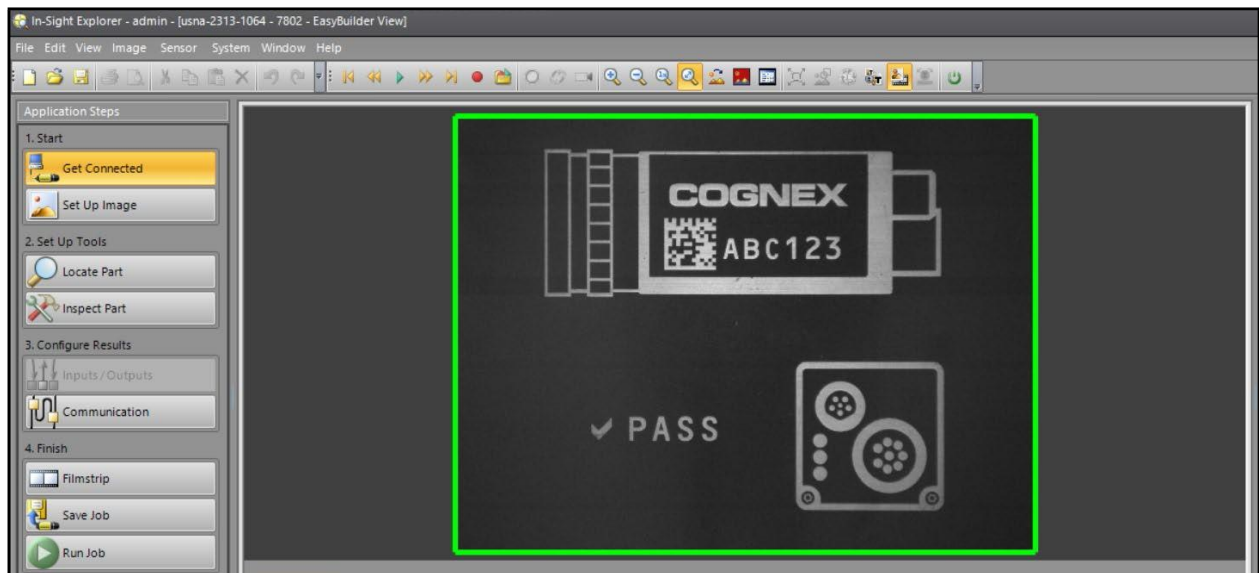
11. Follow the instructions to cycle power on your camera. This will take about 2 minutes, at which point a message will indicate that the reset was successful. Click the **Close** button to close the **Add Sensor** window.
12. Click the **Connect**  button or double click on your camera icon to connect.
13. Confirm you are connected to your camera by checking the following:
 - a. There is a grid pattern attached to your camera in the list.



- b. The top bar of the User Interface (UI) has the camera name / number in it.



- c. There is an image in the center window pane of the UI or the window pane has changed.



NOTE: If you do not have an image, but A and B are verified, you are connected. C will be covered in the next lab.

Lab 1.2 – Getting Connected

At the end of this lab exercise, Participants will be able to:

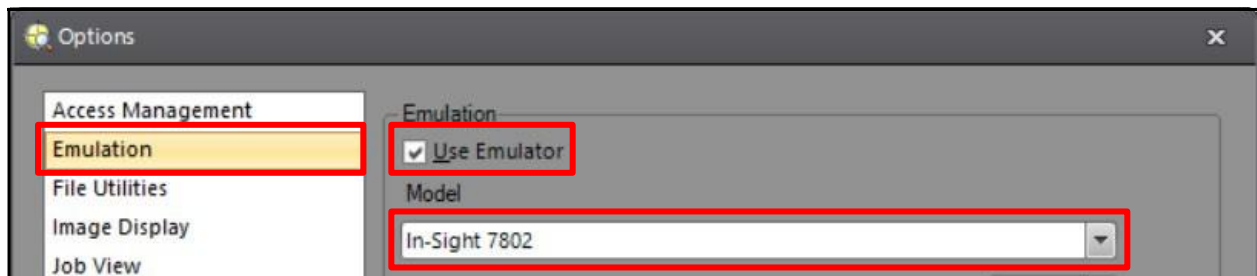
- Emulate the In-Sight 7802

The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

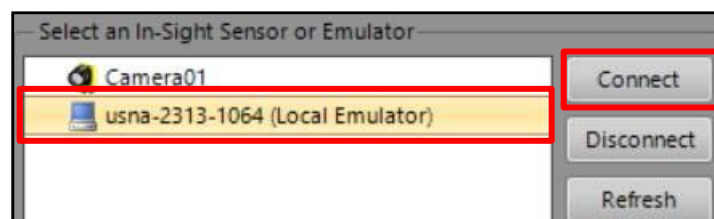
- Options / Emulation

Follow the steps below to complete the lab (using an Emulator):

1. Launch In-Sight Explorer (ISE) and enter the **EasyBuilder** view.
If the EasyBuilder view does not display, from the **View** menu select **EasyBuilder**.
2. Click the **System** menu and select **Options**.
The **Options** dialog displays.



3. Set the parameters as follows:
 - Select **Emulation** from the menu
 - Check the **Use Emulator** checkbox
 - Select **In-Sight 7802** for the ModelClick the **OK** button.
4. If the **Save Current Emulator Job?** dialog displays, click the No button to continue.
5. Select your **Emulator** in the **Select an In-Sight Sensor of Emulator** window and click the **Connect** button.



6. The **Emulator** model will display at the top of the In-Sight Explorer window.

